# Transfer task: plan and begin

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Organize the given measurements and details. Begin your own position prediction.

**SUCCESS CRITERIA - I can...**

1. Organize the task data with fixed axes and matching units.
2. Calculate central and bounded final coordinates.
3. Save an explicit plan for the graph and explanation.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

**LANGUAGE GOAL** Use “component” and “bound” to explain your answer. Sentence starters are optional.

**component**: the part of a vector along one chosen axis

**bound**: an upper or lower limit in a model

**assumption**: a condition accepted for using a model

## 01 Warm-up: bring back an earlier idea

A box at rest has **15 N** right and **3 N** left. **Find the additional horizontal force** needed for balance and **show the sum**.

**Word help:** at rest = not moving in the chosen reference frame

**Sentence starter:** I found ... [unit]. This value means ... in this situation.

**Marzano check - circle:** ① ② ③

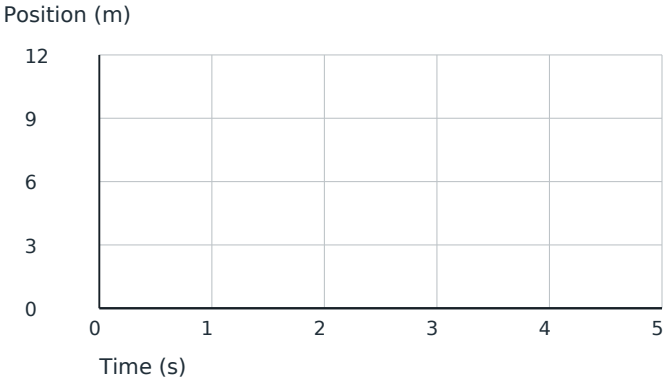
## 02 Worked example: explain the prediction

**Reference:** $x_f = x_i + v_x t$; $y_f = y_i + v_y t$.
Fictional fixed-Earth, constant-velocity calculation. CPA: $x_i$ = **-120,000,000 km**; $y_i$ = **0**; t = **16,000,000 s**; $v_x$ = **8.21 km/s**; $v_y$ = **2.20 km/s**.

**Worked reasoning:** A simplified calculation uses $x_0$ = **0 m** and constant v = **2 m/s** for **5 s**, giving x = **10 m**. This prediction depends on velocity staying constant. It does not prove that the real object moves that way.

**Your turn:** Draw the predicted path and **mark one observation you would** collect to **test the idea** that velocity stays constant.

**Sentence starter:** I predict ... . I used ... to make this prediction because ... .




**Marzano check - circle:** ① ② ③

## 03 Organize the givens

A fictional tracking exercise treats Earth as fixed at the origin. The object moves at constant velocity; gravity and Earth's motion are omitted.
1. Use the given values to **sketch its starting point** and both velocity directions.
2. Label +x toward Earth initially and +y perpendicular to it.

**Sentence starter:** In my work, I used ... to show ... . I labeled it ... .

| Quantity | Value |
|---|---|
| Initial x | -120,000,000 km |
| Initial y | 0 km |
| Time | 16,000,000 s |
| x velocity | +8.21 km/s |
| y velocity | +2.20 km/s |

**Marzano check - circle:** ① ② ③

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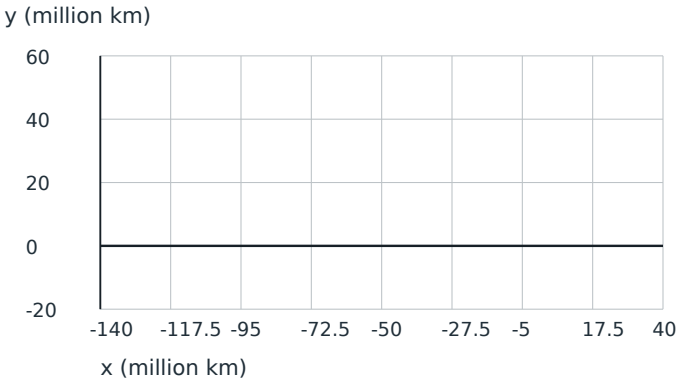
Continue down each column.

Sentence starters are optional. You may plan in any language.

## 04 Find the central end point

Using question **3**, find the change in each coordinate and then the final x and y coordinates. **Show each multiplication** with units. Plot the starting point and final point on the coordinate grid.

**Sentence starter:** I found ... [unit]. This value means ... in this situation.




**Marzano check - circle:** ① ② ③

## 05 Start the bounds

The same calculation uses independent ranges for its starting values: initial x from **-120,050,000** to **-119,950,000 km**; x velocity from **8.02** to **8.40 km/s**; y velocity from **2.15** to **2.25 km/s**. Initial y and time are exact. **Calculate lower** and upper final x and y coordinates.

**Sentence starter:** I found ... [unit]. This value means ... in this situation.

| Coordinate | Lower (km) | Upper (km) |
|---|---|---|
| x | ____ | ____ |
| y | ____ | ____ |

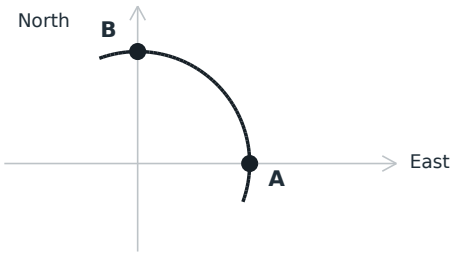
**Marzano check - circle:** ① ② ③

## 06 Review: connect an earlier idea

A rover travels straight *east* at a steady **7 m/s**. A second rover turns at the same steady speed. Which can have zero net force? Explain using velocity.
Use the curve for the turning case; **show how velocity changes**.

**Word help:** net force = the vector sum of all forces on one object

**Sentence starter:** I think ... . I can point to ..., which shows ..., so ... .




Path sketch: runner moves from A toward B.

**Marzano check - circle:** ① ② ③

## 07 Checkpoint

State that your prediction treats velocity as constant. **Name one physical effect it** leaves out. **List what you finished** and your specific next step for Day **22**. Save all work.

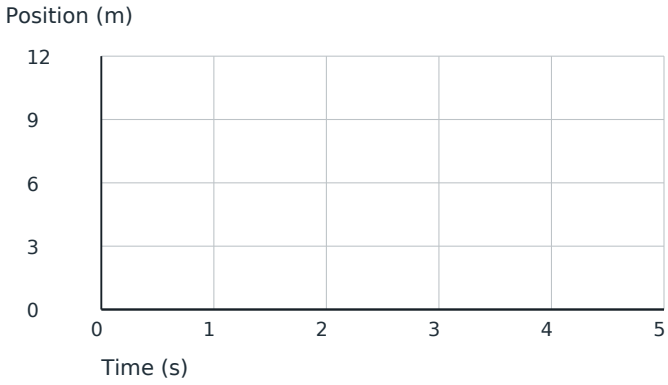
**Sentence starter:** I think ... . I can point to ..., which shows ..., so ... .

**Marzano check - circle:** ① ② ③

## 08 STRETCH: test the prediction

1. Propose two different reasons a measured endpoint could be **8 m**.
2. **Give a measurement** that would help distinguish them.

**Sentence starter:** Both ... . They differ in ... because ... .




**Marzano check - circle:** ① ② ③